TEJ3M Hexadecimal Conversions

Name: _____

Decimal=Base10 →digits? 0-9

Binary=Base2 → digits? 0, 1

Hexadecimal=Base16 →digits? **0–9 and A–F**

Why Hex? Hex compresses binary: 1 hex digit = 4 bits.

Where is Hex used? Memory addresses, MAC addresses, GUIDs/UUIDs, network packet dumps, color codes in web design, file hashes, low-level debugging, assembly/machine code, encoded binary data.

How is Hex represented? Often written with a 0x prefix (0x1A3F) or with subscript 16 ($1A3F_{16}$) or plain 1A3F depending on context.

How do we convert Hex to Decimal? Multiply each hex digit by 16^position and sum.

Example: $2A_{16} = 2\times16^1 + A\times16^0 = 2\times16 + 10 = 32 + 10 = 42$.

How do we convert Decimal to Hex?

Convert the following:

1. 10_{10}=_A____$_{16}$
2. $2A_{16}$=__42___$_{10}$
3. 75_{10}=___4B__$_{16}$
4. $1B_{16}$=___27__$_{10}$
5. 143_{10}=__8F__$_{16}$
6. $7F_{16}$=___127__$_{10}$
7. 35_{10}=__23___$_{16}$
8. $1AF_{16}$=_431___$_{10}$
9. 165_{10}=__A5__$_{16}$
10. 93_{10}=_3A3____$_{16}$

Your computer uses hex to represent all kinds of data. Use the tool at https://onlinetools.com/hex/convert-text-to-hex and type in the Plain Text box the word 'hello' and then try again with 'HELLO'. What is the hex code used to represent the letter h and what hex code is used for H. Note that one of the ways to represent a hex value is to precede it with 0x.

Go to https://htmlcolorcodes.com/ and list the hex codes for the colours RED, GREEN and BLUE.

Computers use hex values to represent their unique address/ID to the internet. Look at the screenshot below and identify the names of this addresses:

```
Connection-specific DNS Suffix  . : EDU.YRDSB.NET
Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . . . . . : B0-4F-13-12-E7-1B
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::1908:f976:1b64:4c74%16(Preferred)
IPv4 Address. . . . . . . . . . . : 10.243.15.2(Preferred)
Subnet Mask . . . . . . . . . . . : 255.255.224.0
Lease Obtained. . . . . . . . . . : Thursday, September 25, 2025 9:06:20 AM
Lease Expires . . . . . . . . . . : Thursday, September 25, 2025 5:06:19 PM
Default Gateway . . . . . . . . . : 10.243.0.2
DHCP Server . . . . . . . . . . . : 10.243.0.50
DHCPv6 IAID . . . . . . . . . . . : 112217875
DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2A-05-DA-4E-B0-4F-13-12-E7-1B
DNS Servers . . . . . . . . . . . : 172.18.0.31
                                    172.18.0.34
                                    172.18.3.3
NetBIOS over Tcpip. . . . . . . . : Enabled
```

Download the zip file called files.zip and extract its contents to your computer. The contents include two text files called file1.txt and file2.txt. Open each file using Notepad. Write down what each file contains below:

file1.txt

file2.txt

Go to the website https://hexed.it/.

Open each of these files using the site. Write down the hex codes for each file below:

file1.txt

file2.txt

Why do you think there is a difference between them?

How big is each file? Why is one larger than the other?

What can you conclude about file data based on the analysis of these two files?

An easier way to determine if two files are identical without actually looking at all of its hex codes and trying to determine if there are differences sis to create a hash value out of the file. A hash value is a unique hexadecimal value for a unique file. If two files, despite appearing to be exactly the same, even being the same size, can still be different files. A different file can be malicious i.e. contain a virus. To find out if two files are identical we can generate a hash value for each one and compare them. Go and select the Tools button for each file at heeded.it and click on the Hash Generator menu option and then generate the hash value. Write them below and compare the values:

file1.txt:

file2.txt:

Download and extract the file from the following zip file: file3.zip. Using the hash generator, determine if the new file called file3.txt is an identical copy of file1.txt or file2.txt.

Play the game at https://flippybitandtheattackofthehexadecimalsfrombase16.com/